Multi-Spectral Imaging

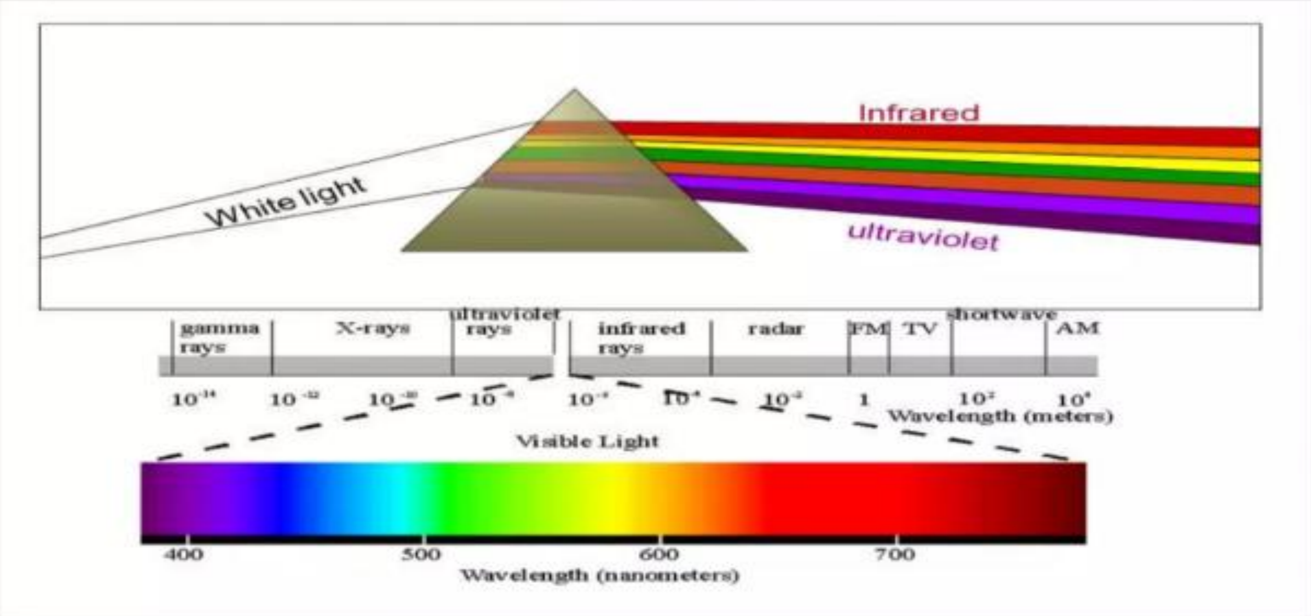
By

Alaa Mohammed Khattab

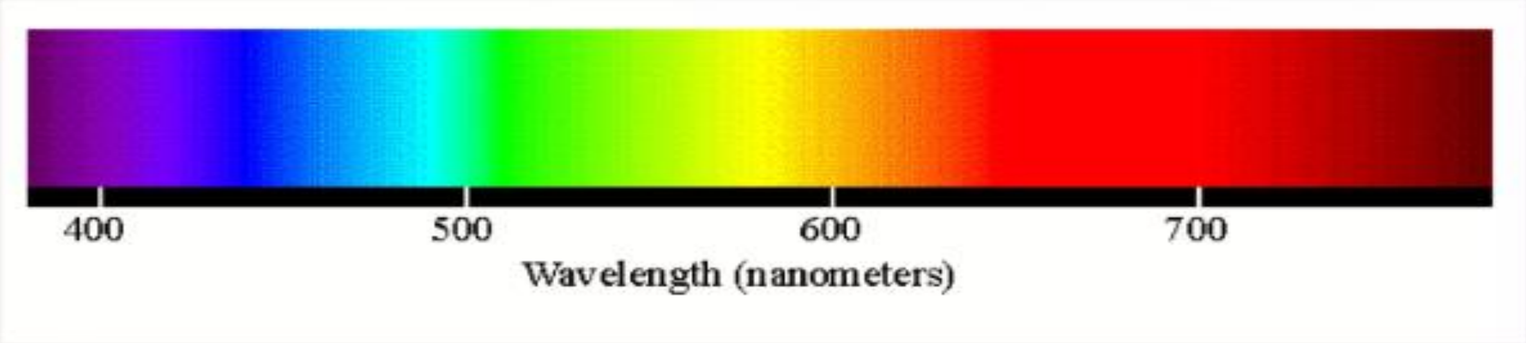
Content

- Introduction.
- Electromagnetic Spectrum.
- Multi-Spectral Imaging.
- How multi-Spectral Imaging can be used.
- Multi-Spectral Cameras.
- Applications.
- Compression.

Electromagnetic Spectrum



Visible Light



- **Blue**: 450-515..520 nm. Used for atmospheric and deep water imaging.
- **Green**: 515..520-590..600 nm. Used for imaging of vegetation and deep water structures.
- **Red**: 600..630-680..690 nm. Used for imaging of man-made objects.

Visible Light

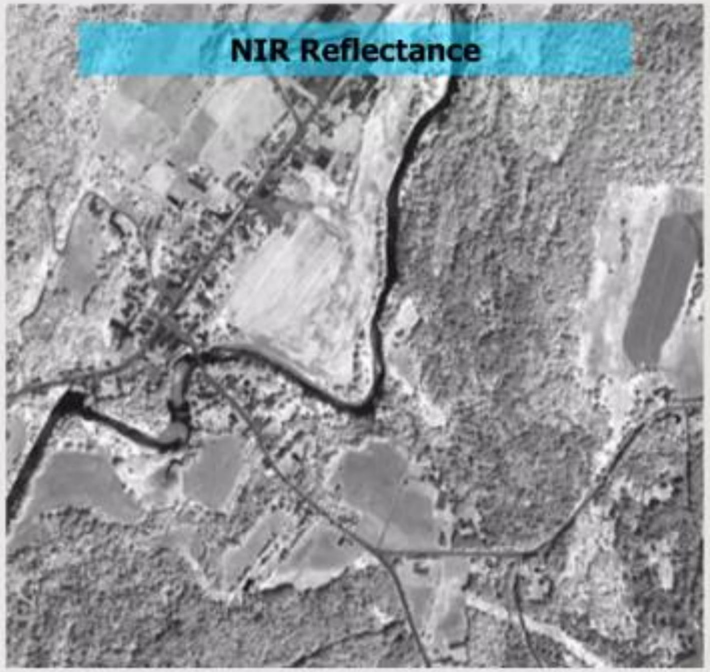
- RGB which lie in the visible range can be combined to produce a full color image for viewing on a display.

- But, RGB are not sufficient to specify the appearance of an object under varying illuminants and viewing conditions.

Infrared

- **Near infrared**: 750-900 nm, is used primarily for imaging of vegetation.
- **Mid-infrared**: 1550-1750 nm, is used for imaging vegetation, soil moisture content, and some forest fires.
- **Far-infrared**: 2080-2350 nm, is used for imaging soil, moisture, geological features, silicates, clays, and fires.

Green vs. NIR



Multi-Spectral Imaging

- A multispectral image is one that captures image data from two or more ranges of frequencies along the spectrum, such as visible light and infrared energy.

- In multispectral images, the same spatial region is captured multiple times using different imaging modalities.

How can be used

- Recording data from the visible spectrum showing an object as it is seen by the eye.

- Take images in area of spectrum infrared region (invisible).

Multispectral Cameras

- Multi-spectral systems make it possible to look beyond those boundaries. Quest innovations is leading in multi-spectral camera technology.

Multispectral Cameras

- Higher resolution pictures due to larger sensors
- Able to see in Wavelengths from 250 – 2200 nm
- Cameras ranging from 1 to 5 channels (a regular camera is 1 channel)

Applications

Medical:

- Multispectral medical images can be formed from different medical imaging modalities such as MRI, CT, and X-ray. These multimodal images are useful for identifying and diagnosing medical disorders.

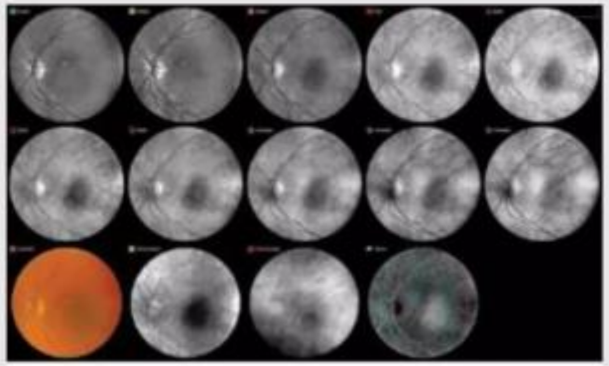


Figure 2. The diagnostic mode of the RHA uses multiple wavelengths of light to capture

Applications

Remote Sensing & Earth Observation

•Land cover classification: Identifies forests, water bodies, agricultural fields, and urban areas.

•Environmental monitoring: Tracks deforestation, desertification, and pollution.

•Climate change studies: Measures glacier retreat, carbon footprint, and ocean temperature variations.

•Disaster response: Aids in flood, wildfire, and earthquake damage assessment.

Example: NASA's Landsat and Sentinel-2 satellites provide high-resolution multispectral images.

Image: Satellite image showcasing different spectral bands used in analysis.

Applications

Agriculture & Precision Farming

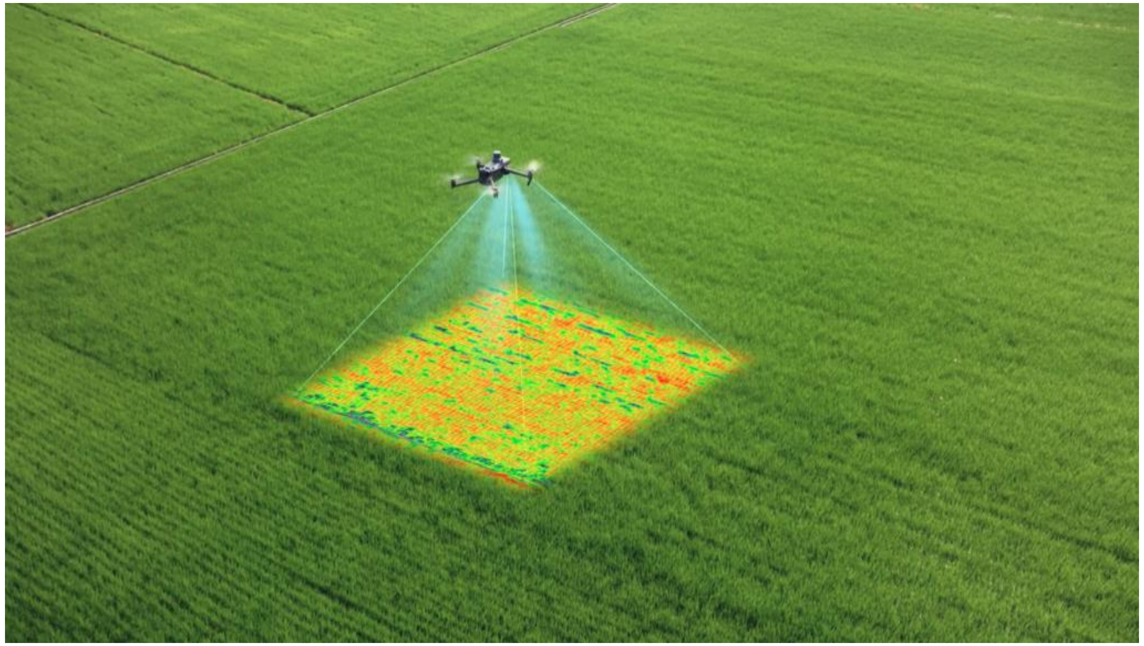
Crop health assessment using NDVI (Normalized Difference Vegetation Index).

Identifying pest infestations and irrigation needs.

Enhancing yield prediction for better resource management.

Example: Drones equipped with multispectral cameras.

Image: NDVI map showing healthy vs. stressed vegetation.



Applications

Medical & Biomedical Imaging

•Early detection of skin diseases, including melanoma and other cancers, using spectral analysis.

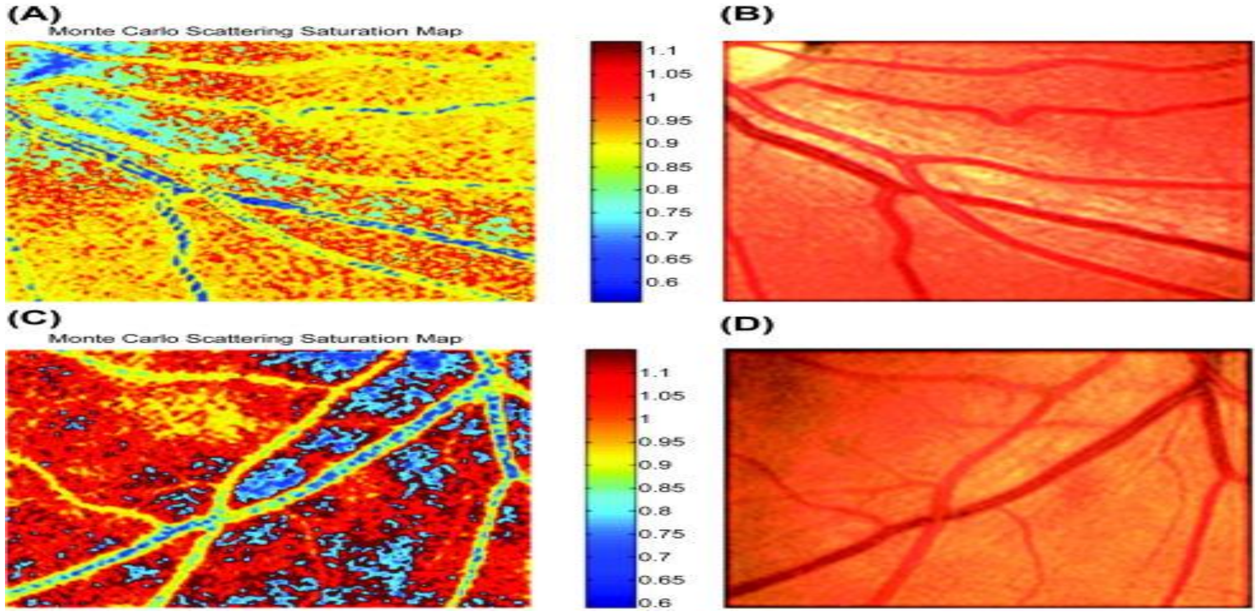
•Blood oxygenation monitoring through near-infrared imaging, critical in medical diagnostics.

•Retinal imaging helps in diagnosing diabetic retinopathy, macular degeneration, and other eye diseases.

•Tissue differentiation in surgeries, improving precision and reducing complications.

Example: MSI is used in endoscopy, dermatology, and surgical guidance for enhanced diagnosis and treatment.

Image: Multispectral scan of human tissue detecting abnormalities.



Applications

Military & Defense

•Enhances surveillance and reconnaissance operations by distinguishing between natural and artificial objects.

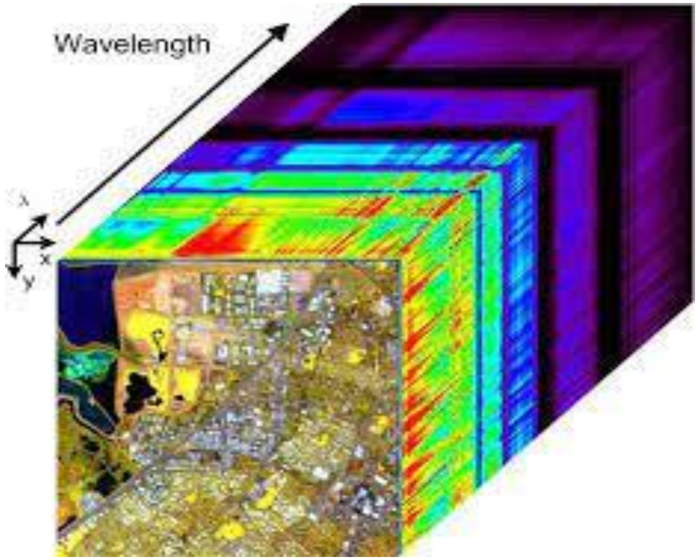
•Helps in camouflage detection by differentiating materials that appear similar to the naked eye.

•Provides night vision and thermal imaging capabilities for battlefield monitoring.

•Assists in missile guidance and target acquisition for improved military strategy.

Example: MSI-equipped UAVs (drones) play a crucial role in real-time intelligence gathering.

Image: Multispectral imagery revealing concealed targets in a combat zone.



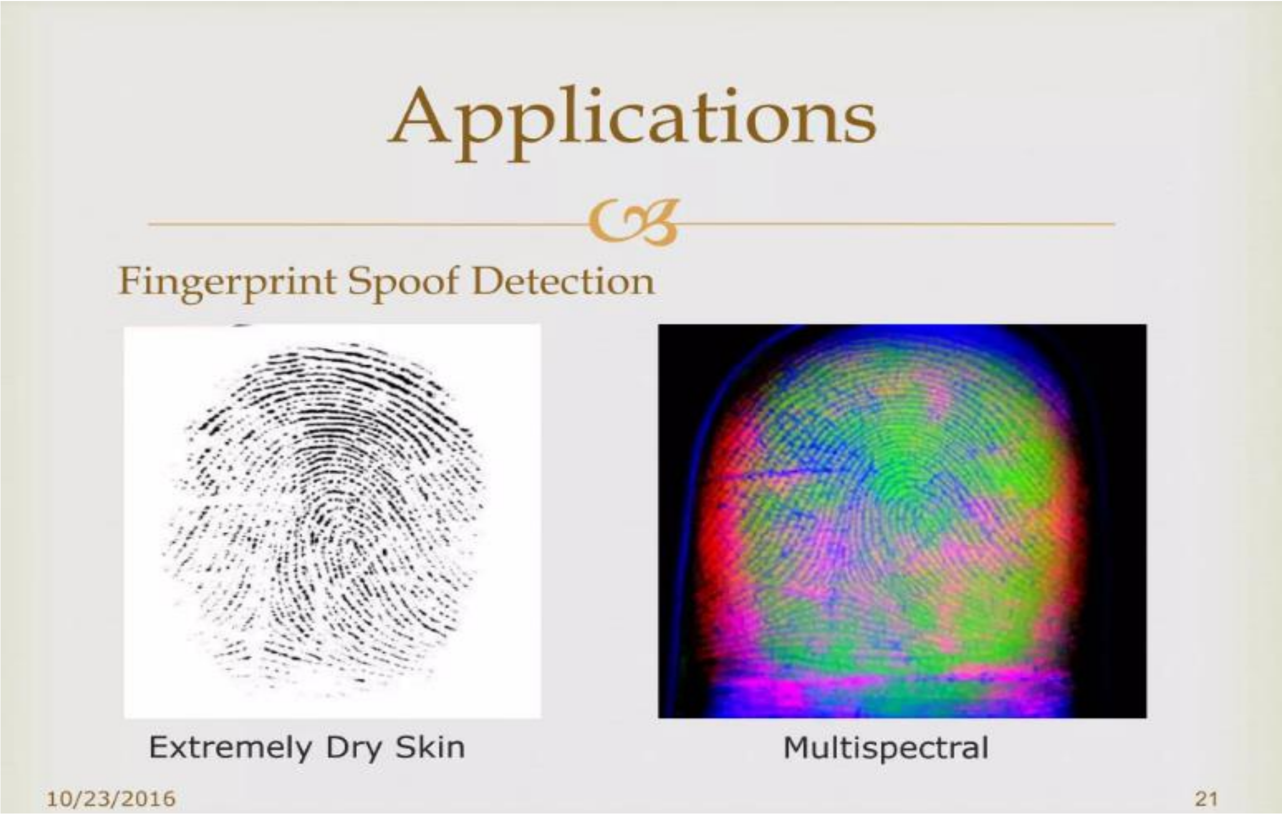
Applications

Fingerprint spoof

Fingerprint spoofing involves creating fake fingerprints to bypass biometric security systems.

•Uses materials like silicone, latex, or gelatin to mimic real fingerprints.

•Fingerprint spoof detection ensures secure and reliable authentication.



Applications

Forest

•Multispectral imaging captures data across different wavelengths, including visible and infrared light.

•Essential for monitoring forests, detecting changes, and assessing environmental impact.

•Used in satellite remote sensing, drone imaging, and ground-based applications.